

- 1) Implement the following functions using (i) 2:1 Mux and NOT gates (ii) 4:1 Mux

(a) $F(A, B, C) = \bar{A}C + BC$

→ (i) using 2:1 Mux & NOT gate
using Select line, $S = C$

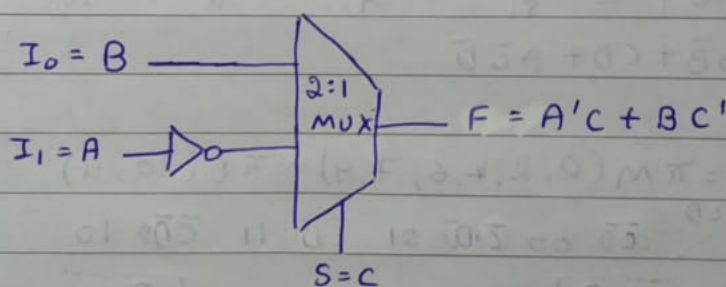
• when $C = 0 \rightarrow I_0 = B$

• when $C = 1 \rightarrow I_1 = A'$

$$\therefore F = C'I_0 + CI_1 = C'(B) + C(A')$$

$$F = A'C + BC'$$

A	B	C	A'	C'	$F = A'C + BC'$
0	0	0	1	1	0
0	0	1	1	0	1
0	1	0	1	1	1
0	1	1	1	0	1
1	0	0	0	1	0
1	0	1	0	0	0
1	1	0	0	1	1
1	1	1	0	0	0



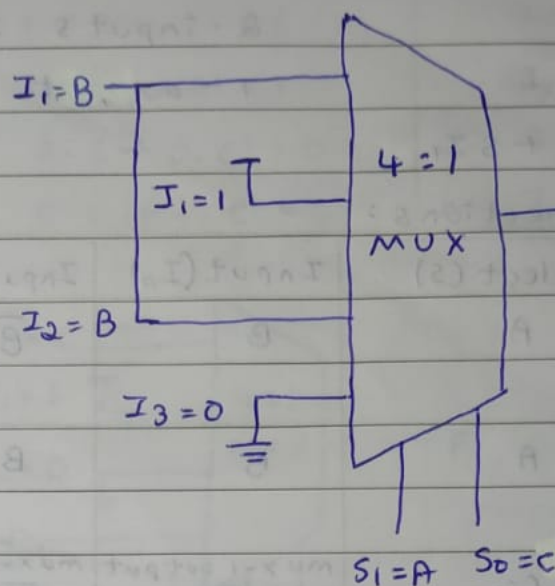
(ii) using 4:1 Mux:

using select lines, $S_1 = A$ $S_0 = C$

The mux inputs are:

A	C	$F = \bar{A}C + BC$	mux input
0	0	$0 + B = B$	$I_0 = B$
0	1	$1 + 0 = 1$	$I_1 = 1$
1	0	$0 + B = B$	$I_2 = B$
1	1	$0 + 0 = 0$	$I_3 = 0$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0



(b) $F(P, Q, R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

→ (i) using 2:1 MUX & NOT gate:

$$\bar{P}\bar{R} + Q\bar{R} + \bar{P}Q(R + \bar{R})$$

$$\bar{P}\bar{R} + Q\bar{R} + \bar{P}Q\bar{R} + \bar{P}QR$$

$$\bar{R}[\bar{P} + Q + \bar{P}Q] + \bar{P}QR$$

w.k.T $A + B + AB = A + B$

$$\bar{R}[\bar{P} + Q] + \bar{P}QR$$

$$R'(P' + Q) + R(P'Q)$$

Here, select line, $S = R$ in MUX-01

- $R = 0 \rightarrow I_0 = P' + Q$

- $R = 1 \rightarrow I_1 = P'Q$

*creating OR function for $I_0 = P' + Q$

here, select line, $S = Q$ in MUX-02

- $Q = 0 \rightarrow$ output should be P'

- $Q = 1 \rightarrow$ output should be 1

$$\therefore \text{output} = Q'(P') + Q(1) = P' + Q$$

*creating AND function for $I_1 = P'Q$

here, select line, $S = Q$ in MUX-03

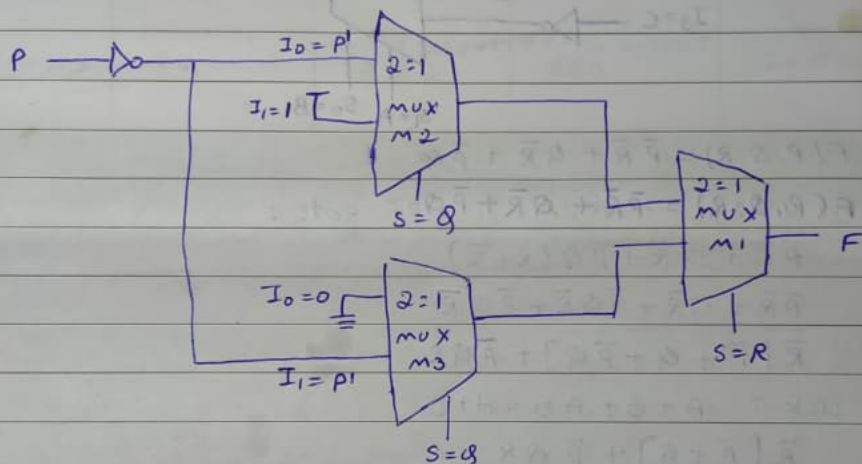
- $Q = 0 \rightarrow$ output should be 0

- $Q = 1 \rightarrow$ output should be P'

$$\therefore \text{output} = Q'(0) + Q(P') = P'Q$$

Truth Table

P	Q	R	P'	R'	$I_0 = P' + Q$	$I_1 = P'Q$	$F = R'I_0 + RI_1$
0	0	0	1	1	1	0	1
0	0	1	1	0	1	0	0
0	1	0	1	1	1	1	1
0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0
1	1	0	0	1	1	0	1
1	1	1	0	0	1	0	0



(iii) using 4:1 MUX:

using select lines, $S_1 = P$ $S_0 = R$

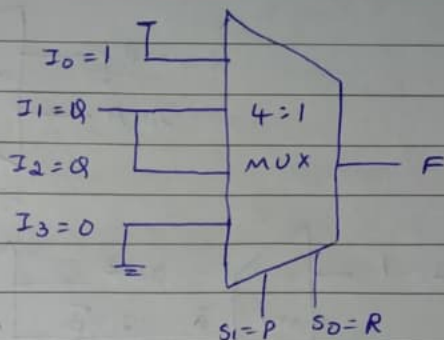
Derivation of MUX inputs

P	R	Substitution	MUX input
0	0	$1 + Q + Q = 1$	$I_0 = 1$
0	1	$0 + 0 + Q = Q$	$I_1 = Q$
1	0	$0 + Q + 0 = Q$	$I_2 = Q$
1	1	$0 + 0 + 0 = 0$	$I_3 = 0$

Truth Table :

P	Q	R	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1

P	Q	R	F
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0



2) Implement the functionality of a full adder using 2:1 Muxes only.

→ The full adder has 3 inputs A, B, cin and gives 2 outputs sum and carry.

$$\text{Sum} = A \oplus B \oplus \text{cin}$$

$$\text{Carry} = AB + (A \oplus B) \text{cin}$$

A 2:1 MUX has - 1 select line (s)

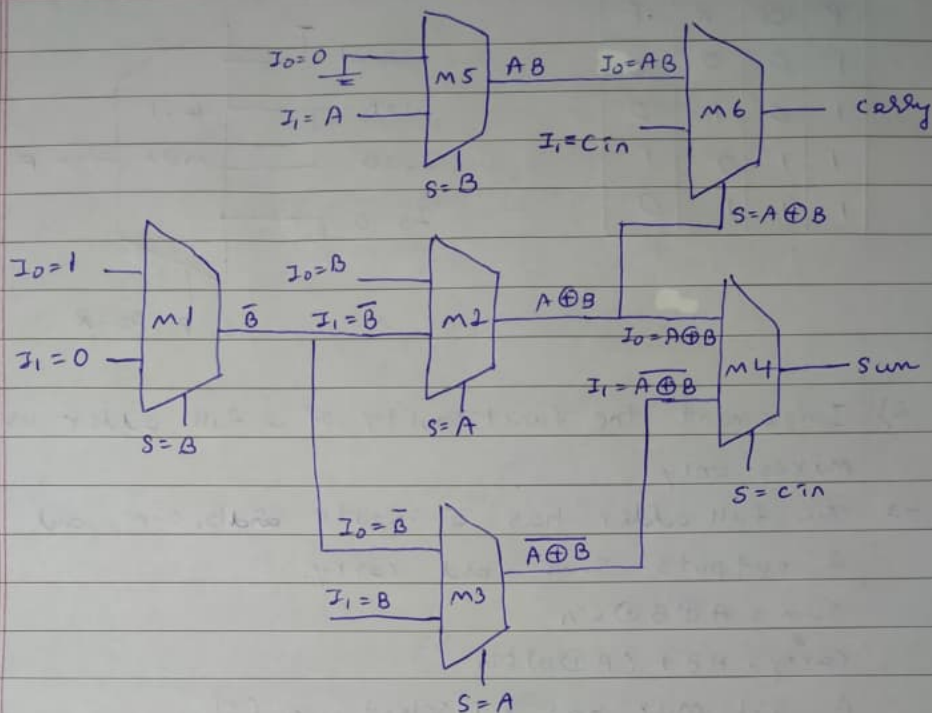
- 2 inputs (I_0 & I_1)

- 1 output (Y)

$$Y = \bar{S} I_0 + S I_1$$

MUX Connections:

MUX	select (s)	I/P (I_0)	I/P (I_1)	output
MUX-1	B	1	0	$\bar{B}$
MUX-2	A	B	MUX-1 o/p $\bar{B}$	$A \oplus B$
MUX-3	A	MUX-1 o/p $\bar{B}$	B	$\overline{(A \oplus B)}$
MUX-4	cin	MUX-2 o/p $A \oplus B$	MUX-3 o/p $\overline{(A \oplus B)}$	sum
MUX-5	B	0	A	AB
MUX-6	MUX-2 o/p $A \oplus B$	MUX-5 o/p AB	cin	carry



- 3) simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$
 and implement the SOP expression using 4=1 MUX
 $\rightarrow f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$
 by using K-Map,

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
00 $\bar{A}\bar{B}$				1
01 $\bar{A}B$	0	1	3	2
11 AB	4	5	7	6
10 $A\bar{B}$	X	X	X	X
00 $\bar{A}\bar{B}$				1
01 $\bar{A}B$				
11 AB				
10 $A\bar{B}$				

hence, the simplified SOP expression is:
 $\therefore F = B\bar{C} + C\bar{D}$

In a 4:1 MUX, There are 4 inputs I_0, I_1, I_2, I_3 and 2 select lines S_0 & S_1 and 1 output.

• $S_1 = C$

• $S_0 = D$

C	D	F	MUX input
0	0	$B(1) + 0(1) = B$	$I_0 = B$
0	1	$B(1) + 0(0) = B$	$I_1 = B$
1	0	$B(0) + 1(1) = 1$	$I_2 = 1$
1	1	$B(0) + 1(0) = 0$	$I_3 = 0$

